

- 13** An object that undergoes simple harmonic motion has an amplitude A and total energy E .

What is the displacement of the object from the equilibrium position when its kinetic energy is $\frac{3}{4} E$?

A 0.25 A

B 0.50 A

C 0.75 A

D 0.87 A

B When $x = 0$, $E_k = E$

$$E = \frac{1}{2} m \omega^2 A^2$$

$$E_k = \frac{1}{2} m \omega^2 (A^2 - x^2)$$

$$\frac{3}{4} E = \frac{1}{2} m \omega^2 (A^2 - x^2)$$

$$\frac{3}{4} (\frac{1}{2} m \omega^2 A^2) = \frac{1}{2} m \omega^2 (A^2 - x^2)$$

$$\frac{3}{4} A^2 = (A^2 - x^2)$$

$$x = 0.50 A$$

14 A beam of unpolarised light passes through three polaroids P_1 , P_2 and P_3 . The polarising